

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

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INTRODUCTION

Importance

Chronic Obstructive Pulmonary Disease (COPD) is a heterogeneous disorder characterised by airflow obstruction that is not fully reversible. The airflow limitation is usually both progressive and associated with exposure to noxious particles or gases. Smoking is by far the most important risk factor.

Although not fully reversible, COPD is preventable and clinically manageable even within primary care. Main treatment goals in managing stable COPD are reducing symptoms and risk of future exacerbations. Managing it with appropriate inhalers, as well as non-drug measures, are important in reducing COPD-associated morbidity and mortality.

Epidemiology

Globally, in 2019, COPD is the third most common cause of death and an increasingly important contributor to morbidity due to an ageing population, urbanisation, and persistence of risk factors. In Singapore, COPD is estimated to be the tenth highest cause of death and seventeenth highest cause of disability-adjusted life years. Locally, COPD contributes to chronic respiratory disease-related hospitalisations and emergency department visits and was estimated to have an annual societal cost of SGD\$3,304 per capita in 2022. COPD exacerbations account for the greatest proportion of the total COPD burden on the healthcare system.

SCREENING AND DIAGNOSIS

Screening in the Community

- Screen all patients with any of the risk factors found in [Table 1](#) for COPD symptoms, and vice versa, at least yearly
- ***Suspect COPD in any patient with at least one COPD symptom and risk factor.***
- ***All patients who are suspected to have COPD based on symptoms MUST BE evaluated by spirometry***
- Screening spirometry in the general asymptomatic population is not recommended

COPD Symptoms, Risk Factors, Co-Morbidities

Table 1 - Key indicators for considering a diagnosis of COPD

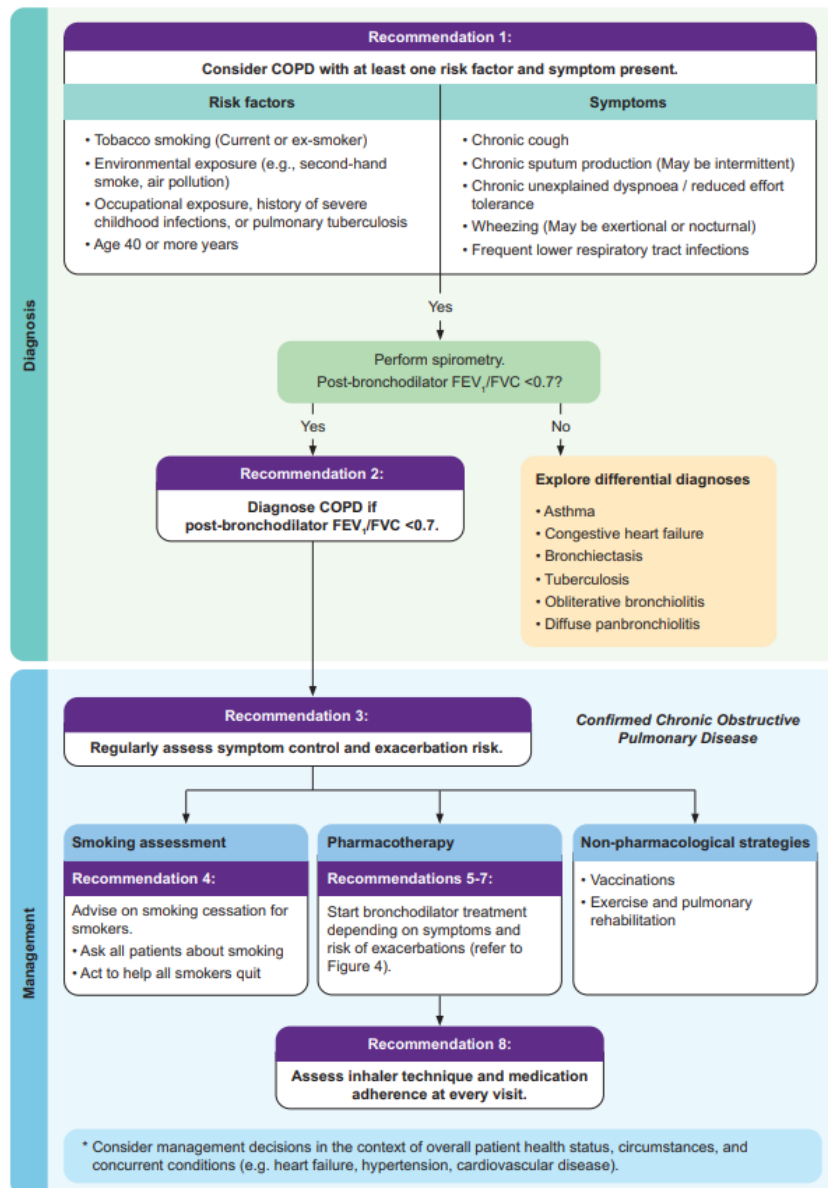
COPD SYMPTOMS	RISK FACTORS	CO-MORBIDITIES
Chronic cough • Generally, the initial symptom • May be intermittent and may be unproductive Chronic sputum productions (any pattern, may be intermittent) Chronic unexplained dyspnoea or reduced effort tolerance • COPD's hallmark • Progressive over time • Persistent Recurrent lower respiratory tract infections Wheezing (may be exertional or nocturnal) Fatigue Additional clinical features in severe COPD: weight loss, muscle mass loss, anorexia, ankle swelling (cor pulmonale), and depression / anxiety	• Age 40 years and above • Tobacco smoke: ex and current smoker • Environmental exposure: second-hand smoke, air pollution • Occupational exposure: dust, vapor, fumes, gases, and other chemicals • History of abnormal lung development, severe childhood infections, or pulmonary tuberculosis • Rare risk factor: alpha-1-antitrypsin deficiency	• Heart disease • Hypertension • Diabetes • Chronic kidney disease • Osteoporosis • Sleep apnoea • Depression • Cognitive impairment • Lung cancer

Diagnosis of COPD

Requires all of the following *to be met*:

- At least one COPD symptom
- At least one risk factor
- Evidence of airflow limitation: **post-bronchodilator** spirometry $FEV_1 / FVC < 0.70$

Figure 1. Overview of Diagnosis and Management of COPD
 (Adapted from ACE Clinical Guidelines COPD - Diagnosis and Management Dec 2024)



FEV_1/FVC , Ratio between the forced expiratory volume in one second (FEV_1) and forced vital capacity (FVC)

Table 2 - Summary of clinical presentation and diagnostic evaluation of COPD

HISTORY
Risk factors • Family history of COPD or other chronic respiratory disease • Smoking history: age at initiation, average amount smoked per day since initiation, date when stopped or a current smoker • Environmental exposure: household or outdoor, chronologically taken • History of childhood pulmonary infections, HIV, or Tuberculosis • Asthma, allergy, sinusitis, or nasal polyps • Possibilities for reducing risk factors, especially smoking cessation
Presence of comorbidities: heart disease, osteoporosis, musculoskeletal disorders, anxiety and depression, and malignancies
Symptoms (try to elicit the pattern of symptom development, frequency and intensity of symptoms): • Dyspnoea: ask about amount of effort required to induce uncomfortable breathing • Cough • Wheezing (indicates presence of airflow obstruction) • Acute chest illness: ask about occurrence and frequency of episodes of increased cough and sputum with wheezing, dyspnoea, and fever
Use of reliever medications
History of exacerbations or previous hospitalization for respiratory disorders: Increased risk of future exacerbation if they had: • TWO OR MORE exacerbations requiring antibiotics or steroids in the previous year OR • ONE exacerbation leading to hospitalization in the previous year
Impact of disease on patient's life: activity limitation, missed work and economic impact, effects on family routines, feelings of depression or anxiety, well-being
Social and family support available to the patients
PHYSICAL EXAMINATION (generally present only with severe disease)
Chest: • Emphysema (only when severe): overdistension of the lungs in stable state (chest held near full inspiratory position at end of normal expiration, low diaphragmatic position), decreased intensity of breath and heart sounds, and prolonged expiratory phase • Evidence of airflow obstruction: wheezes during auscultation on slow or forced breathing and prolongation of forced expiratory time • Frequently observed with severe disease (characteristic, but not diagnostic): pursed-lip breathing, use of accessory respiratory muscles, retraction of lower interspaces
Others: • Unusual positions to relieve dyspnoea at rest • Digital clubbing is not typical in COPD (even with associated hypoxemia) and suggests other diagnoses (e.g., lung cancer, bronchiectasis, pulmonary fibrosis, cirrhosis, inflammatory bowel diseases) • Mild dependent oedema may be seen in the absence of right heart failure

INVESTIGATION
Spirometry: • Mandatory to establish COPD diagnosis (post-bronchodilator FEV₁ / FVC < 0.70 confirms COPD) • May be performed at pulmonary function laboratories or in clinics using HSA-approved portable office spirometers using the same post-bronchodilator testing criteria • If an HSA-approved portable office spirometer is used, ensure it can calculate the post-bronchodilator FEV₁/FVC • Should be undertaken when patients are clinically stable and free from respiratory tract infections
Pulse oximetry: to evaluate the need for supplemental oxygen therapy
Chest Xray: • Not useful to establish COPD diagnosis • Valuable to exclude alternative diagnosis and establish comorbidities (concomitant respiratory, skeletal, and cardiac diseases) • CXR changes associated with COPD: signs of lung hyperinflation, hyperlucency of the lungs, rapid tapering of vascular markings Full blood count: • To rule out anaemia • Blood eosinophil count to predict the magnitude of the effect of inhaled corticosteroids (ICS) in preventing future exacerbation: ○ Blood eosinophil count < 100 cells/μl: ICS-containing regimens have little or no effect ○ Blood eosinophil count 100-299 cells/μl: can consider adding ICS if symptoms not better ○ Blood eosinophil count ≥ 300 cells/μl: ICS-containing regimens are beneficial
Other investigations to consider: • Alpha-1 antitrypsin deficiency (AATD) screening: WHO recommends that all COPD patients should be screened once especially in areas with high AATD prevalence. A low concentration (< 20% normal) is highly suggestive of homozygous deficiency. Family members should also be screened • CT chest: not routinely recommended except for detection of bronchiectasis and COPD patients that meet the criteria for lung cancer risk assessment • Lung volumes and diffusing capacity • Exercise testing and assessment of physical activity

Differential Diagnosis of COPD

Table 3 - Differential diagnosis of COPD

COPD		Asthma	
• Age of onset: 40 or more years of age • Pattern of respiratory symptoms: ○ Symptoms persist despite treatment ○ Days with stable and unstable symptoms consistent daily symptoms and exertional dyspnoea ○ Chronic cough or sputum may precede dyspnoea, unrelated to trigger • History, family history, or risk factors: ○ Previously diagnosed with COPD, chronic bronchitis, or emphysema by a doctor ○ Exposure to risk factors such as tobacco smoke or history of severe childhood infections or pulmonary tuberculosis • Time course: ○ Symptoms slowly worsening over time (progressive course over years) ○ Rapid-acting bronchodilator treatment provides only transient relief • Lung function: abnormal • Lung function between symptoms: ○ Record of post-bronchodilator $FEV_1/FVC < 0.7$ ○ Persistent expiratory airflow limitation • CXR: hyperinflated lung fields (in some patients) [this finding is not required for diagnosis]		• Age of onset: <40 years, but can manifest at any age • Pattern of respiratory symptoms: ○ Symptoms vary widely from day to day ○ Symptoms worse at night or early morning ○ Symptoms often triggered by exercise, temperature change, dust, or allergen exposure • History, family history, or risk factors: ○ Previously diagnosed with asthma by a doctor ○ History of other allergic conditions (for example, allergic rhinitis, eczema, allergic conjunctivitis, or childhood wheeze) ○ Family history of asthma and other allergic conditions (for example, allergic rhinitis or eczema) • Time course: ○ Symptoms do not worsen progressively; they vary seasonally, or from year to year ○ May improve spontaneously or have an immediate response to bronchodilator or to inhaled corticosteroids over weeks • Lung function: ○ Record of variable airflow limitation (spirometry, peak flow) ○ Persistent expiratory airflow limitation may be present • Lung function between symptoms: often normal • CXR: normal	
Tuberculosis (TB)	Obliterative Bronchiolitis	Diffuse Panbronchiolitis	
• Onset all ages • Chronic cough • Hemoptysis • CXR shows lung infiltrates • Microbiological confirmation • High local prevalence of TB	• Onset at younger age, non-smokers • May have history of rheumatoid arthritis or acute fume exposure • Seen after lung or bone marrow transplantation • HRCT on expiration shows hypodense areas	• Predominantly seen in patients of Asian descent • Most patients are male and non-smokers • Almost all have chronic sinusitis • CXR and HRCT shows diffuse small centrilobular nodular opacities and hyperinflation	

*** If a concurrent diagnosis of asthma is suspected, the pharmacotherapy options should follow asthma guidelines. If the distinction between COPD and asthma is unclear, consider a trial of asthma treatment first**

Other Causes of Chronic Cough

Table 4 - Other causes of chronic cough

INTRATHORACIC	EXTRATHORACIC
Asthma Lung Cancer Tuberculosis Bronchiectasis Left Heart Failure Interstitial Lung Disease Cystic Fibrosis Idiopathic Cough	Chronic Allergic Rhinitis Upper Airway Cough Syndrome (UACS) Gastroesophageal Reflux Medications (e.g., ACE inhibitors)

MANAGEMENT

Goals of Management

1. **To reduce symptoms:** relieve symptoms, improve exercise tolerance, and improve health status
2. **To reduce risks:** prevent disease progression, prevent, and treat exacerbation, and reduce mortality
3. **To prevent or minimize side effects from treatment**

Refer to [Figure. 1](#) for overview of diagnosis and management of COPD

Follow-Up and Monitoring

1. **Assess symptoms at least yearly**, and more frequently for patients who are more symptomatic, have more frequent exacerbations, or have recent escalation in treatment:

- Check for: **frequency and intensity of symptoms, use of reliever medications, and impact on activities of daily living**
- **Use CAT score.** Document in EPIC “Flowsheets”

Figure 2. COPD Assessment Test (CAT): CAT preferred over mMRC scale since the latter does not account for other COPD symptoms
(Adapted from GOLD 2025 guidelines)

CAT™ Assessment

For each item below, place a mark (x) in the box that best describes you currently.
Be sure to only select one response for each question.

EXAMPLE: I am very happy	0	1	2	3	4	5	I am very sad	Score
I never cough	0	1	2	3	4	5	I cough all the time	
I have no phlegm (mucus) in my chest at all	0	1	2	3	4	5	My chest is completely full of phlegm (mucus)	
My chest does not feel tight at all	0	1	2	3	4	5	My chest feels very tight	
When I walk up a hill or one flight of stairs I am not breathless	0	1	2	3	4	5	When I walk up a hill or one flight of stairs I am very breathless	
I am not limited doing any activities at home	0	1	2	3	4	5	I am very limited doing activities at home	
I am confident leaving my home despite my lung condition	0	1	2	3	4	5	I am not at all confident leaving my home because of my lung condition	
I sleep soundly	0	1	2	3	4	5	I don't sleep soundly because of my lung condition	
I have lots of energy	0	1	2	3	4	5	I have no energy at all	

Reference: Jones et al. ERJ 2009; 34 (3): 648-54.

TOTAL SCORE:

2. **History of exacerbations** (informs the patient's risk of future exacerbation):

- Increased risk of future exacerbation if they have had:
 - **TWO OR MORE COPD exacerbations** requiring antibiotics or steroids in the previous year OR
 - **ONE exacerbation leading to hospitalisation** in the previous year
- Definition of COPD exacerbation: an event characterised by dyspnoea and/or cough and sputum that worsens in <14 days which may be accompanied by tachypnoea and/or tachycardia and is often associated with increased local and systemic inflammation caused by infection, pollution, or other insults to the airways

3. Smoking assessment (2 As approach):

- Ask all patients about smoking
- Act to help all smokers quit (can refer for smoking cessation services)
- Refer to NUP Smoking Cessation CPG for further details

4. Pharmacotherapy: optimise bronchodilator treatment and assess inhaler technique and medication adherence at every visit (*refer to Figure. 4*)

5. Manage other co-morbidities

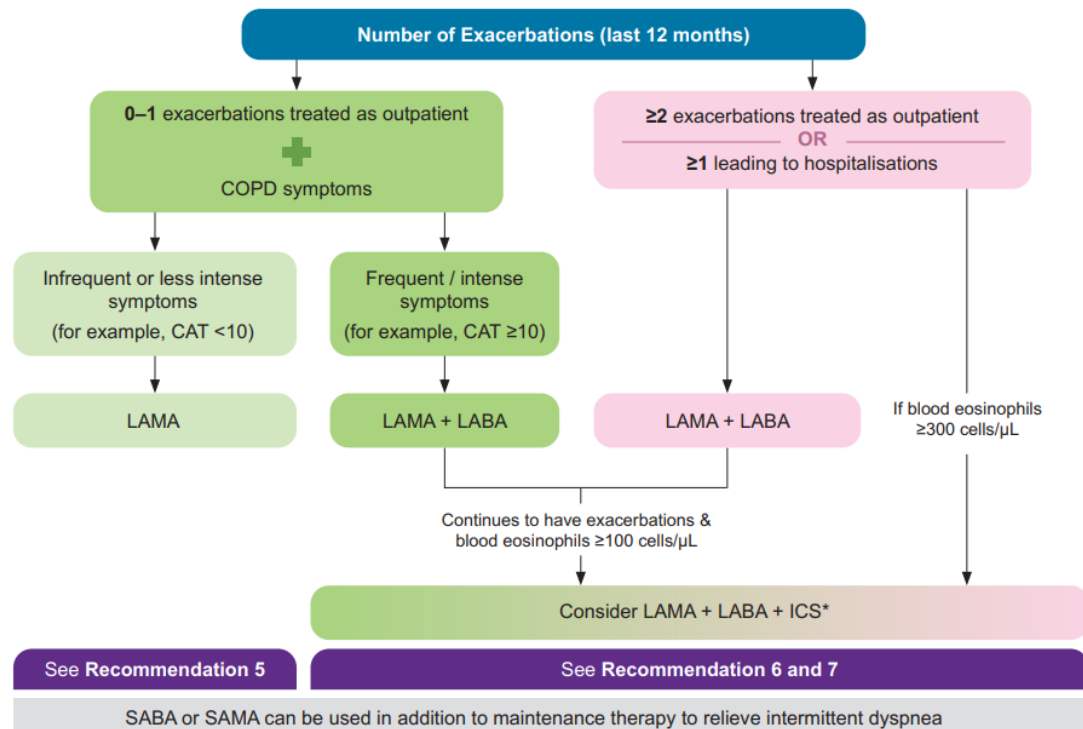
6. Non-pharmacological strategies:

Ensure up-to-date vaccination
• Annual influenza vaccine • Pneumococcal vaccine (<u>refer to NUP detailed recommendation on pneumococcal vaccines</u>) • Tdap vaccine for those who have not received vaccination against Pertussis • Additional vaccinations recommended by GOLD 2025 guidelines: ○ Covid 19 vaccine based on WHO and CDC recommendation ○ Respiratory Syncytial Virus (RSV) vaccinations for individuals aged >60 years and/or with chronic heart or lung disease ○ Zoster vaccine for COPD patient over 50 years to protect against Shingles
Encourage exercise: aerobic exercise 3-4x/week for 20-30 minutes and strengthening exercises
Assess the need for pulmonary rehabilitation (to be done at restructured hospital)
○ Consider in patients with high symptom burden and risk of exacerbations ○ Includes structured exercise training (twice a week for 6-8 weeks), education, instruction in various respiratory and chest physiotherapy techniques, and psychosocial support
Assess the need for long term oxygen therapy (LTOT)
○ Indications for LTOT (>15 hours/day) when: ❖ $\text{SaO}_2 < 88\%$ on room air when stable (confirmed 2x over a 3-week period) OR ❖ $\text{SaO}_2 = 88\%$ on room air with evidence of right heart failure or erythrocytosis ○ If prescribed, LTOT should achieve $\text{SaO}_2 \geq 90\%$, and clinicians should review their patient's condition and SaO_2 at room air every 60 to 90 days to adjust the oxygen therapy accordingly ○ Inappropriate use of O_2 therapy may result in respiratory depression ○ Patients must be given adequate instructions about the source of O_2 to be used, method of delivery, duration of use, and flow rates to be used for different levels of physical activity (e.g. at rest, during exercise, during sleep) ○ Patients should be warned to discontinue smoking when prescribed O_2
Nutritional support
○ Weight loss and malnutrition may develop as COPD progresses and this indicates a poor prognosis ○ In such case, nutritional repletion (including protein supplementation) plays an important role for such patients, and should be coupled with optimization of lung function, regular exercise, and oxygenation if needed

Pharmacotherapy

Figure 3. Individualized maintenance pharmacotherapy options for COPD patients

(Adapted from ACG COPD Guidelines Dec 2024)



CAT, COPD Assessment Test (CATTM); ICS, inhaled corticosteroid; LABA, long-acting beta2-agonist; LAMA long-acting muscarinic antagonist; SABA, short-acting beta2-agonist; SAMA, short-acting muscarinic antagonist.

* Consider adjusting down the dose or withdrawing ICS if pneumonia or other associated side effects of ICS occur. But caution in patients with blood eosinophils ≥300 cells/μL as this can be associated with the development of exacerbations.

† LABA-only inhalers are not available in Singapore at the time of ACG publication; see Recommendation 5 for more details.

Other medications for management of COPD:

- Methylxanthines (e.g. theophylline), PDE4 inhibitors, mucolytics, and macrolides are not within the scope of this CPG
- They should be reserved as adjuncts to inhaled therapy

Recommendation 5 (ACG COPD guidelines):

- Start a long-acting bronchodilator treatment, **preferably a LAMA** (preferred over LABA-only inhalers), for patients with infrequent or less intense symptoms and lower risk of exacerbation
- **SAMA or SABA alone** can be considered in patients with very occasional dyspnea
- **SABA or SAMA can be used as PRN** to relieve intermittent dyspnea in all COPD patients.

Recommendation 6 (ACG COPD guidelines):

- Start dual bronchodilator therapy with LAMA+LABA for patients with frequent or intense COPD symptoms, or a higher risk of exacerbation

Recommendation 7 (ACG COPD guidelines):

- Consider screening FBC for eosinophilia (at least 14 days after COPD exacerbation) before starting ICS in COPD patients
- Consider a triple therapy with LAMA+LABA+ICS for patients with:
 1. Higher risk for COPD exacerbations and have blood eosinophils ≥ 300 cells/μl
 2. Frequent COPD exacerbations on LAMA+LABA therapy and blood eosinophils ≥ 100 cells/ μl
 3. History of asthma or for patients with features of both asthma and COPD: consider specialist referral for this group of patients
- Consider using single inhaler triple therapy (SITT) instead to simplify treatment (SITT not available in NUP as of September 2025)
- Side effects of ICS: oral thrush, hoarse voice, skin bruising, and pneumonia
- Avoid ICS in patients with recurrent pneumonia, blood eosinophils < 100 cells/μl, or history of mycobacterial infections

Figure 4. Inhalers for COPD registered in Singapore (Adapted from ACG COPD Guidelines Dec 2024)

Relievers – short-acting bronchodilators			
SABA		SAMA	SAMA + SABA
Salbutamol (Salbuair MDI*, Azmasol MDI*, Ventolin Evohaler MDI, Buventol Easyhaler DPI*) 		Ipratropium (Iprovent MDI*) 	Ipratropium + fenoterol (Berodual N MDI) 
Maintenance – long-acting bronchodilators			
LAMA			
Umeclidinium (Incruse Ellipta DPI) 	Glycopyrronium (Seebri Breezhaler DPI) 	Tiotropium (Spiriva Respimat SMI) 	
LAMA + LABA			
Umeclidinium + vilanterol (Anoro Ellipta DPI) 	Glycopyrronium + indacaterol (Ultibro Breezhaler DPI) 	Tiotropium + olodaterol (Spiolto Respimat SMI) 	
Maintenance – ICS [†] combinations			
LABA + ICS			
Formoterol + budesonide (Symbicort Turbuhaler DPI [‡] , Symbicort Rapihaler MDI [‡] , DuoResp Spiromax DPI*) 	Salmeterol + fluticasone propionate (Seretide Accuhaler DPI [‡] , Salfumix Easyhaler DPI [‡]) 	Vilanterol + fluticasone furoate (Relvar Ellipta DPI [‡]) 	Formoterol + beclomethasone dipropionate (Foster MDI, Foster NEXThaler DPI) 
LAMA + LABA + ICS			
Umeclidinium + vilanterol + fluticasone furoate (Trelegy Ellipta DPI [‡]) 	Glycopyrronium + formoterol + beclomethasone dipropionate (Trimbow MDI) 	Glycopyrronium + formoterol + budesonide (Breztri Aerosphere MDI) 	
Check if your patient is doubling up on inhalers			
Do not double up on inhalers from the same class of drugs. For example, if a patient is already on one LAMA such as umeclidinium, do not prescribe another, such as glycopyrronium or tiotropium.	Do not double up on inhalers containing a muscarinic antagonist (SAMA, LAMA, or LAMA + LABA). This includes a SAMA with LAMA, because of potential antimuscarinic or anticholinergic side effects such as dry mouth and urinary retention.	Do not double up on inhalers containing a LABA (LAMA + LABA, LABA + ICS). Common side effects include tremors, palpitations, and headaches.	

List of inhalers currently registered and available for COPD management in Singapore. Active ingredients in bold denote availability on government subsidy list. Please refer to product inserts for detailed information on the inhalers.

* Generics.

[†] ICS (beclomethasone, budesonide, fluticasone) should only be used in combination with LAMA+LABA.

[‡] Not all strengths available are registered for COPD.

Symbicort Turbuhaler: only 9/320 mcg and 4.5/160 mcg
 Symbicort Rapihaler: only the 2.25/80 mcg and 4.5/160 mcg

Seretide Accuhaler: only 50/500 mcg
 Salfumix Easyhaler: only 50/500 mcg

Relvar Ellipta: only 25/100 mcg
 Trelegy Ellipta: only 100/62.5/25 mcg

DPI, dry powder inhaler; ICS, inhaled corticosteroid; LABA, long-acting beta₂-agonist; LAMA long-acting muscarinic antagonist; mcg, microgram; MDI, metered dose inhaler; SABA, short-acting beta₂-agonist; SAMA, short-acting muscarinic antagonist; SMI, soft mist inhaler

Commonly Used COPD Medications (Please refer to [NUP Intranet Asthma / COPD Medication Chart](#) for more information)

Table 5 - Commonly used COPD medications in Singapore

COPD Medications	Adult Dose	Significant Adverse Reactions	Contraindications
Short acting bronchodilators Recommended only for patients with very occasional dyspnoea OR used as PRN to relieve intermittent dyspnoea in all COPD patients			
SABA: salbutamol (Ventolin) 100mcg 	MDI: 1-2 puffs 3-4 times PRN	Hypersensitivity reactions (e.g. urticaria, angioedema, rash, bronchospasm, oropharyngeal oedema) Hypokalaemia (high doses)	Hypersensitivity to salbutamol or any component of the formulation; severe hypersensitivity to milk proteins (dry powder inhalers).
SAMA: ipratropium bromide (Atrovent N) 20mcg 	MDI: 2 puffs 3-4 times PRN	Dry mouth, constipation, tachycardia, palpitations, arrhythmias, nausea and vomiting, dyspepsia, headaches, dizziness, ocular complications (e.g. mydriasis, narrow-angle glaucoma)	Hypersensitivity to ipratropium, atropine, or its derivatives

COPD Medications	Adult Dose	Significant Adverse Reactions	Contraindications
Long-acting bronchodilators: preferred over short acting bronchodilators and if for monotherapy, LAMA alone is preferred over LABA alone			
LAMA: umeclidinium bromide (Incruse Ellipta) 	DPI (62.5mcg): 1 INH OD. Max 1 INH/day	CV effects (e.g. cardiac arrhythmia, atrial fibrillation, tachycardia), hypersensitivity reactions (e.g. anaphylaxis, angioedema, pruritus, rash, urticaria); increased intraocular pressure, exacerbated urinary retention	Hypersensitivity to umeclidinium or any component of the formulation; severe hypersensitivity to milk proteins
LAMA: tiotropium bromide monohydrate (Spiriva) 	Spiriva Respimat soft mist inhaler (2.5mcg): 2 INH OD. Max 2 INH/day	Xerostomia, URTI, pharyngitis, sinusitis	Hypersensitivity to ipratropium, tiotropium, or any component of the formulation
LAMA: Glycopyrronium bromide (Seebri Breezhaler) (Not available in NUP) 	DPI (50 mcg per capsule): Contents of 1 capsule inhaled OD using Breezhaler device. Max 1 capsule inhaled/day	Drowsiness, blurred vision, constipation, intestinal pseudo-obstruction, heat prostration, diarrhoea, incomplete intestinal obstruction, urinary retention, bronchospasm, hypersensitivity reactions	Hypersensitivity to glycopyrrolate or any component of the formulation. Severe hypersensitivity to milk proteins.

COPD Medications	Adult Dose	Significant Adverse Reactions	Contraindications
LABA+ LAMA combination: preferred for patients with frequent or intense COPD symptoms, or a higher risk of exacerbation			
LABA+ LAMA: vilanterol+ umeclidinium (Anoro Ellipta) 	DPI (25/62.5mcg): 1 INH OD. Max 1 INH/day	Hypersensitivity reactions (e.g. anaphylaxis, angioedema, rash, urticaria), blood pressure and tachycardia, atrial fibrillation, hyperglycaemia, diabetes mellitus and ketoacidosis exacerbation; hypokalaemia, urinary retention, narrow-angle glaucoma, bladder outlet obstruction, CNS stimulation/excitation; thyrotoxicosis exacerbation.	Hypersensitivity to umeclidinium, vilanterol, or any component of the formulation Severe hypersensitivity to milk proteins Asthma monotherapy Acute episodes of bronchospasm or acutely deteriorating COPD Concomitant use of other LABA
LABA+ LAMA: olodaterol+ tiotropium bromide (Spiolto Respimat) (Not available in NUP) 	Spiolto Respimat Soft mist inhaler (2.5/2.5mcg): 2 INH OD. Max 2INH/day	Nasopharyngitis, back pain, cough	Hypersensitivity to tiotropium, ipratropium, olodaterol, atropine or any component of the formulation Asthma monotherapy
LABA+ LAMA: indacaterol+ glycopyrronium (Ultibro Breezhaler) (Not available in NUP) 	DPI (110/50mcg per capsule): Contents of 1 capsule inhaled OD using Breezhaler device. Max 1 capsule inhaled/day	CNS depression, hypersensitivity reactions (e.g. angioedema, urticaria), hypertension, increased heart rate, ECG changes, hyperglycaemia, diabetes mellitus and ketoacidosis exacerbation, thyroid stimulation, hypokalaemia, narrow-angle glaucoma, urinary retention	Hypersensitivity to indacaterol, glycopyrrolate, or any component of the formulation Severe hypersensitivity to milk proteins Asthma monotherapy

COPD Medications	Adult Dose	Significant Adverse Reactions	Contraindications
ICS combinations - Long term monotherapy with ICS is not recommended - LABA+LAMA+ICS is superior to LABA+ICS combination in COPD management - In patients with COPD + asthma features, treatment should always contain an ICS			
LABA+ LAMA+ ICS: Vilanterol/umeclidinium/Fluticasone (Trelegy Ellipta) Registered for COPD: only Trelegy 25/62.5/100mcg (Not available in NUP) 	DPI (25/62.5/100mcg): 1 INH OD. Max 1 INH/day	Nasopharyngitis, headache, constipation, diarrhoea, dysgeusia, oral candidiasis, UTI, influenza, pneumonia, bronchitis, URTI	Hypersensitivity to fluticasone, umeclidinium, vilanterol, or any component of the formulation Severe hypersensitivity to milk proteins Primary treatment of status asthmaticus or other acute episodes of COPD or asthma where intensive measures are required
LABA+ ICS: vilanterol+ fluticasone (Relvar Ellipta) Registered for COPD: only Relva 25/100mcg 	DPI (25/100mcg): 1 INH OD. Max 1 INH/day	Hypertension, extrasystoles, headache, voice disorder, oropharyngeal candidiasis, upper abdominal pain, nasopharyngitis, URTI, arthralgia	Hypersensitivity to fluticasone, vilanterol, or any component of the formulation Severe hypersensitivity to milk protein Primary treatment of status asthmaticus or other acute episodes of COPD or asthma where intensive measures are required
LABA+ ICS: formoterol+ budesonide Registered for COPD: Duoresp Spiromax 4.5/160mcg Symbicort Turbuhaler: only 4.5/160mcg and 9/320mcg (Not available in NUP): Symbicort Rapihaler: only 2.25/80mcg and 4.5/160mcg (Not available in NUP)	DPI DUORESP (Spiromax) (4.5/160mcg): 2 INH BD (Max dose) DPI Symbicort Turbuhaler (4.5/160mcg): 2 INH BD (Max dose) MDI Symbicort Rapihaler (2.25/80mcg): 4 INH BD (Max dose)	Headache, nasopharyngitis, URTI, pneumonia, abdominal distress, oral candidiasis, skin bruises	Hypersensitivity to budesonide, formoterol, or any component of the formulation Primary treatment of status asthmaticus or other acute episodes of asthma or COPD where intensive measures are required.

COPD Medications	Adult Dose	Significant Adverse Reactions	Contraindications
			
LABA+ ICS: salmeterol+ fluticasone (Seretide) <i>Registered for COPD: only Seretide Accuhaler 50/500mcg formulation</i> 	DPI Seretide Accuhaler (50/500mcg): 1 INH BD (Max dose)	Hypokalaemia, paradoxical bronchospasm, increased wheezing, shortness of breath, QTc interval prolongation	Hypersensitivity to fluticasone, salmeterol, or any component of the formulation Status asthmaticus Acute episodes of asthma or COPD Severe hypersensitivity to milk proteins
Others			
Methylxanthines: - Not recommended unless other long-term bronchodilators are unavailable or unaffordable - Toxicity is dose-related and most of the benefit occurs only when near-toxic doses are given - Have significant interactions with commonly used medications such as erythromycin, ciprofloxacin, allopurinol, cimetidine, and fluvoxamine	Individualise dose based on serum theophylline levels (use ideal body weight to calculate dose) Theophylline (anhydrous) MR tab/cap: 125mg to 250mg BD Theophylline (monohydrate) MR tab (Not available in NUP): 200mg BD (up to 300mg BD or 400mg BD) Elderly: dose reduction may be necessary	Atrial and ventricular arrhythmia Grand mal convulsions Urinary retention (particularly in elderly males) Other side effects: headaches, insomnia, nausea, and heartburn	Porphyria, recent MI, acute tachyarrhythmia Concomitant use with ephedrine (in children < 6 years or with bodyweight < 22 kg).

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of COPD, including health promotion and prevention / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Coordinator

- Perform general screening (i.e. fall risk, social economics, smoking & drinking history) and offer smoking cessation clinic
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal)

3. Care Manager

- Assess the learning needs and barriers of the patient and caregiver.
- Evaluate the level of understanding on chronic obstructive lung disease and lifestyle measures.
- Identify reasons for suboptimal/poor adherence to treatment plans and devise strategies to motivate the patients.
- Can do CAT score to assess severity of COPD
- Provide education on medications (inhalers) and inhaler technique
- Make referral to MSW, pharmacist and psychologist if needed

4. Advanced Practice Nurse

- Manage patients with COPD within scope of practice
- Initiate and titrate medication within the scope of APN practice
- Offer timely influenza and pneumococcal vaccinations
- Encourage smoking cessation

5. Dietitian

- Dietary management for patients with malnutrition / underweight, or poor appetite / oral intake, for oral supplementation
- Dietary management for patients who are overweight / obese

6. Medical Social Worker

- Advance Care Planning for patients with severe COPD
- Psychosocial assessment, consideration, and support

7. Physiotherapist

- To assess patient's physical fitness for appropriate exercise prescription
- To educate, motivate and provide advice on appropriate physical activities
- To teach positions to ease shortness of breath and secretion clearance techniques if indicated

8. Psychologist

- Psychological and behavioural interventions to manage psychological stress, improve disease management and quality of life.
- Psychological assessment and intervention to manage co-occurring psychological problems (depression, anxiety disorders *etc.*)

9. Pharmacist

- Smoking cessation clinic
- Detect, prevent and address drug-drug / drug-disease interactions
- Perform medication reconciliation

SPECIAL SITUATIONS

Acute Exacerbation of COPD

Definition of COPD exacerbation: an event characterized by dyspnoea and/or cough and sputum production that worsens over ≤ 14 days; **may be accompanied by tachypnoea and/or tachycardia**

Risk factors:

- History of prior exacerbations: best predictor
- Others: advanced age, longer duration of COPD, chronic bronchitis, elevated Eosinophils, more severe dyspnoea or symptoms, severe airflow obstruction, use of medications associated with sedation or respiratory depression, and comorbid disease (especially heart disease, GERD, and pulmonary hypertension)

Triggers for exacerbation:

- Respiratory infections, **most commonly viral or bacterial-** account for 70% of COPD exacerbation
- Other triggers: environmental pollution, weather conditions, pulmonary embolism, or an unknown aetiology

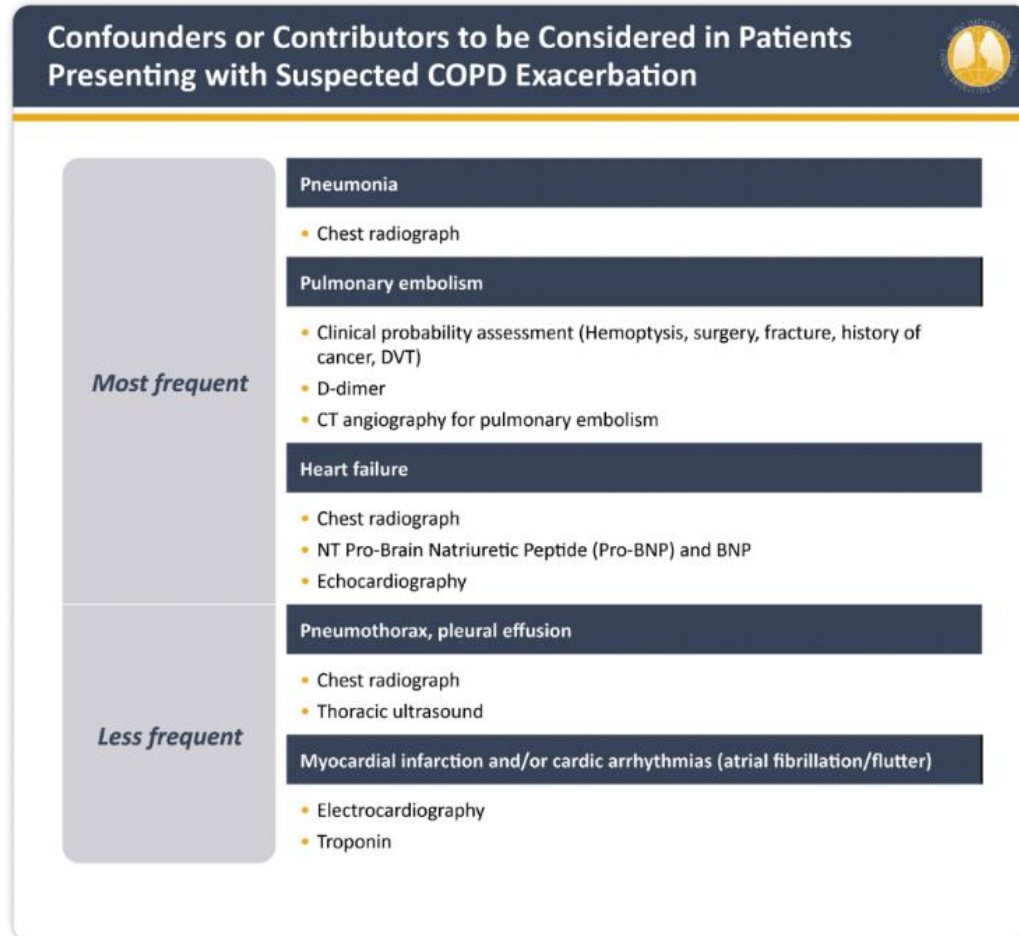
Symptoms: 3 cardinal symptoms of dyspnoea, cough, and/or sputum production; develop over several hours to days

Physical examination findings:

- Wheezing and tachypnoea
- Features of respiratory compromise: use of accessory muscles, fragmented speech, orthopnoea, and depressed mental status

Differential diagnosis of COPD exacerbation: pneumonia, pulmonary embolism, congestive heart failure, cardiac arrhythmia, acute coronary syndrome, pneumothorax, and pleural effusion

Figure 5. Differential diagnosis of COPD exacerbation (Adapted from GOLD 2025 guidelines)



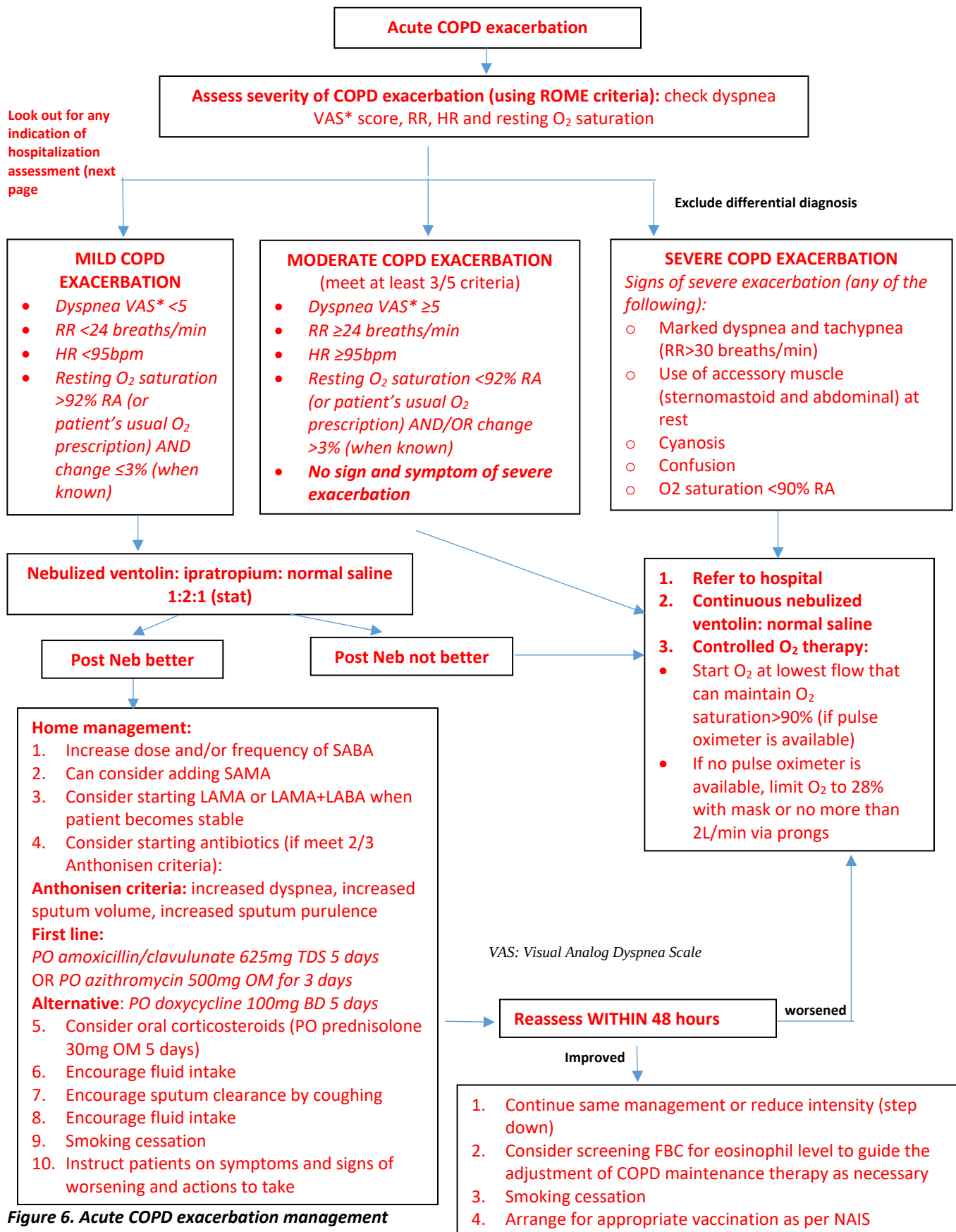


Figure 6. Acute COPD exacerbation management

Indication for hospitalization assessment:

- Moderate to severe exacerbation
- Acute respiratory failure
- Onset of new physical signs (e.g. cyanosis, peripheral oedema)
- Failure of an exacerbation to respond to initial medical management
- Presence of serious comorbidities (e.g. heart failure, newly occurring arrhythmia, *etc.*)
- Insufficient home support

REFERRALS AND COMMUNITY RESOURCES

Indication for Respiratory Medicine Referral

- Severe or frequent exacerbations: suspected severe COPD, frequent infections or recurrent cough or wheeze
- Difficult or complicated COPD:
 - Onset of cor pulmonale
 - Bullous lung disease: to assess for surgery
 - Assessment for LTOT
 - Home nebuliser therapy
- Uncertain diagnosis:
 - Disease with age < 40 years and < 10 pack years (1 pack year = 20 cigarettes/day for 1 year): TRO AATD
 - Rapid decline in FEV₁ (> 60mL/year decrease in FEV₁)
 - Symptoms disproportionate to FEV₁
 - Development of new symptoms such as haemoptysis or new physical signs

Palliative and Supportive Care

- Palliative treatment options to reduce dyspnoea includes opioids, pulmonary rehabilitation, patient self-management education for breathing techniques, neuromuscular electrical stimulation, chest wall vibration, and blowing air onto the face – in addition to treatment with inhalers
- Consider referral to palliative care services to aid in managing refractory dyspnoea
- Consider performing ADVANCED CARE PLANNING (ACP) early: recent hospitalization is a good opportunity to start the discussion

RECOMMENDED CARE COMPONENTS

Recommended care components	Minimum frequency	Remarks
Weight and BMI [^]	Yearly	
CAT score [^]	Yearly	
Spirometry	For diagnosis	
Smoking assessment [^]	Yearly for smokers Once-off for non-smokers, unless there is a change in a smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provide smoking cessation counselling
Inhaler technique	Every follow-up and prior to modifying current therapy	
Influenza Vaccination [^]	Yearly	
Pneumococcal Vaccination	Ensure up to date according to NAIS	
Covid 19 Vaccination	Ensure up to date following National Guidelines	

[^]Reportable clinical indicators

ABBREVIATION

BMI: Body Mass Index	LABA: Long-acting B ₂ -agonists
CAT: COPD Assessment Test	LAMA: Long-acting muscarinic antagonists
COPD: Chronic Obstructive Pulmonary Disease	mMRC: modified Medical Research Council
CT: Computer Tomography	NAIS: National Adult Immunisation Schedule
CXR: Chest X Ray	PDE4: Phosphodiesterase-4
FEV ₁ : Forced Expiratory Volume in 1 second	RSV: Respiratory Syncytial Virus
FVC: Forced Vital Capacity	SABA: Short-acting B ₂ -agonists
ICS: Inhaled corticosteroids	SAMA: Short-acting muscarinic antagonists

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